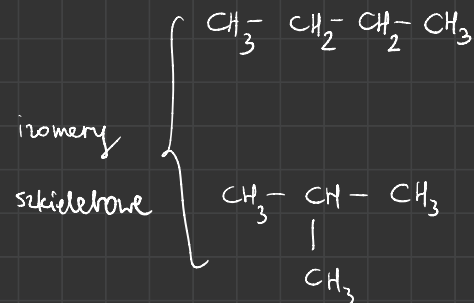
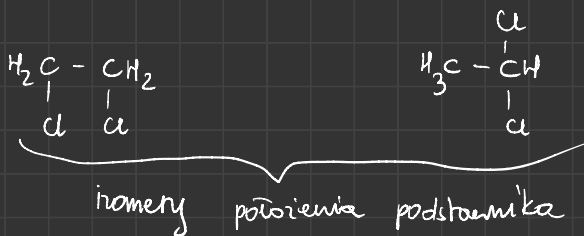


C_4H_{10} - węglowodór nasycony

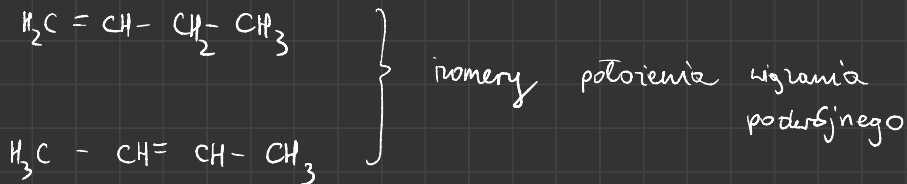


wzory półstrukturalne

$C_2H_4Cl_2$



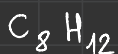
C_4H_8



Węglowodory $\rightarrow$ tylko C i H

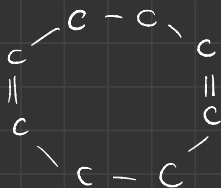
$\rightarrow$ bez wiązań wielokrotnych i pierścieni: alkanany $C_n H_{n+2}$

Stopień nienasyżenia: $\frac{[\text{liczba atomów H brakujących w stosunku do alkanany}]}{2}$
(SN)

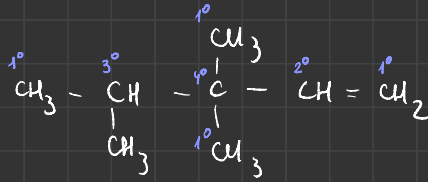


gdyby był alkanem: $C_8 H_{18}$

$$SN = \frac{18-12}{2} = 3$$

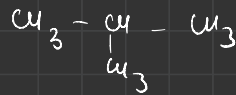


Rzędowość atomu węgla - 2 iloma at. węgle jest połączony innymi atomami węgla

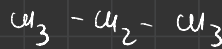
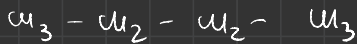


Szeregi homologiczne

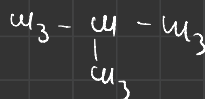
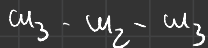
- alkanany $C_n H_{2n+2}$
- alkeny $C_n H_{2n}$ (mają wiązanie podwójne)
- alkiny $C_n H_{2n-2}$ (mają wiązanie potrójne)
- cykloalkany $C_n H_{2n}$ (mają pierścień)



izomery

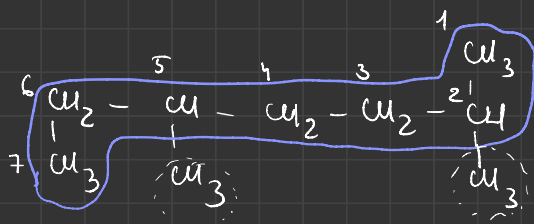


homologi

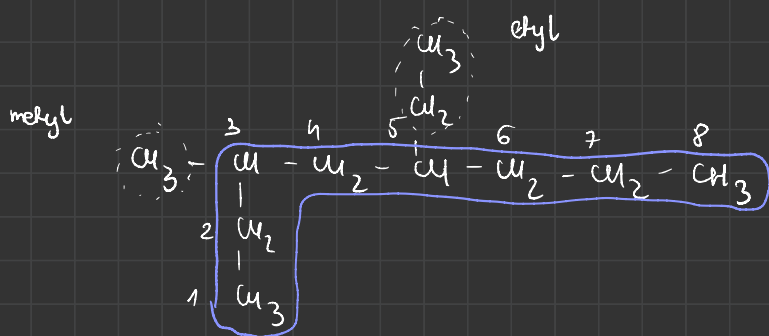


ani izomery,
ani homologi

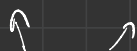
Nazewnictwo alkanów



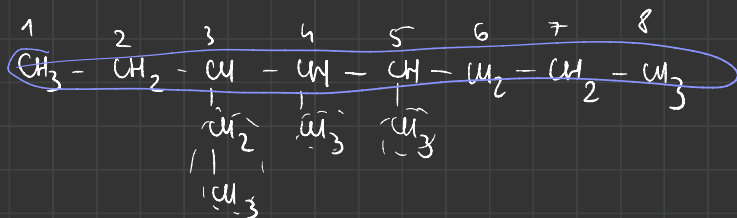
2,5 - dimetyloheptan



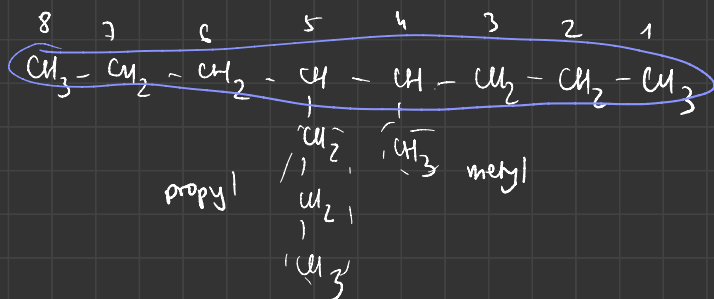
5 - etyl - 3 - metyloktan



w kolejności alfabetycznej (di/mi/tebra nie uwzględniamy)

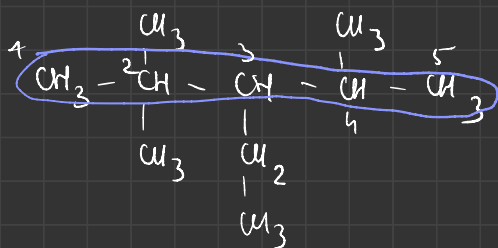


3 - etyl - 4,5 - dimetyloktan



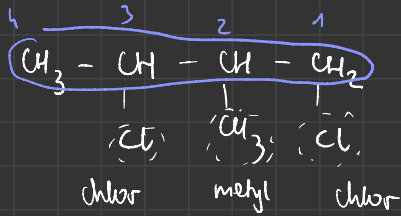
jeśli są te same numery z obu stron - wybieramy ten, aby pierwszy alfabetycznie miał mniejszy numer

4-metylo-5-propyloktan

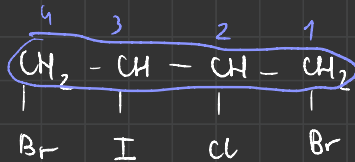


3-etylo-2,2,4-trimetylpentan

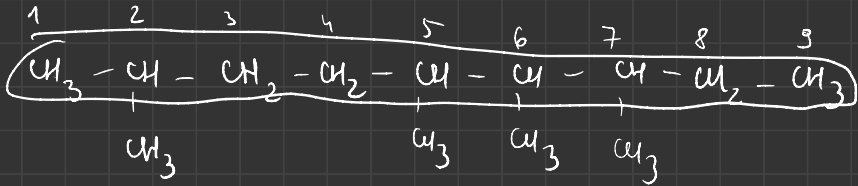
Nazewnictwo polimerych alkanów



1,3-dichloro-2-metylobutan

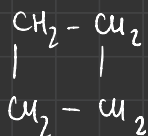


1,4-dibromo-2-chloro-3-jodobutan



2,5,6,7-tetrametylononan

Naznamišite cykloalkane (i podobne)



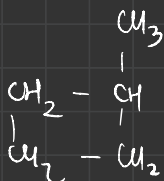
$\equiv$



cyklobutan



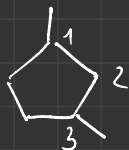
cyklopentan



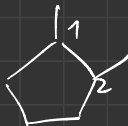
$\equiv$



metylcyklobutan



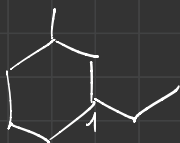
1,3 - dimetylcyklopentan



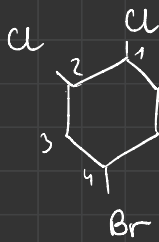
1,2 - dimetylcyklopentan



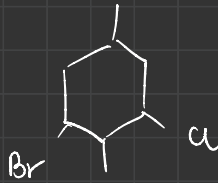
etylcykloheksan



1-etyl-3-metylcykloheksan



4-bromo - 1,2 - dichlorocykloheksan



1-bromo - 3-chloro - 2,5-dimetylo

cykloheksan